



Can a carbon emission trading scheme generate the Porter effect? Evidence from pilot areas in China

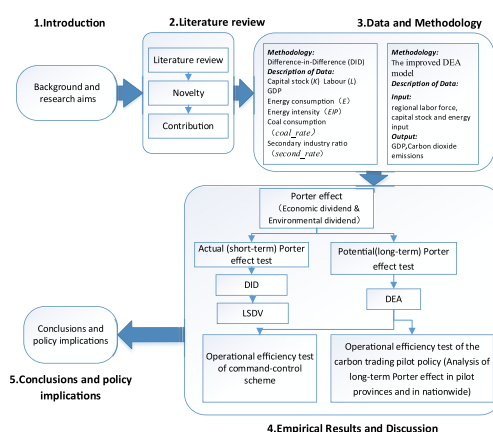
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HIGHLIGHTS

- The difference-in-difference (DID) method and the improved DEA model are employed.
- ETS does not realize the Porter effect in the short term.
- In the long term ETS can achieve the Porter effect.
- ETS has good economic and emission reduction functions.

GRAPHICAL ABSTRACT



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ABSTRACT

“Kyoto Protocol” states that carbon market is one of three private emission-reduction instruments. Since 2013 China implemented carbon emission trading scheme (ETS) in seven provinces or cities. The purpose of this paper is to investigate whether the implementation of ETS can result in the Porter effect. Based on the Porter hypothesis theory, this study employs the difference-in-difference (DID) method and the improved DEA model to analyze whether ETS can bring economic dividend and environmental dividend. The empirical results specify that in the short term, ETS can significantly reduce carbon emissions in the pilot provinces, but fail to increase GDP. Therefore, ETS does not realize the Porter effect in the short term. Nevertheless, in terms of the empirical results we can find ETS plays a significant role in emission reduction. In the long term, ETS can stimulate sustainable economic dividend and environmental dividend, and achieve the Porter effect. From the test results, we can find ETS has good economic and emission reduction functions. ETS achieves the Porter effect in the long term but not in the short term. In order to achieve the Porter effect from ETS successfully, a sound carbon emission trading scheme must be established to ensure efficient carbon emission trading market.

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1. Introduction

Since the 19th National Congress of the Communist Party of China, protection of the ecological environment has become the center of attention again (Dong et al., 2019). The development goals for the

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ecological civilization and beautification of China were put forward by Chinese government. In December 2017, the National Development and Reform Commission (NDRC) announced that a carbon emission trading market for the power generation industry was being officially launched nationwide. It signifies that the establishment of a carbon emission trading market is on track in China. Prompted by climate change, this initiative not only reflects China's determination for carbon emission reduction, but also positively begins a series of commitments to reduce carbon emissions by practical action at the international level. Since China's reform and opening up, the country's economy has enjoyed a period of high development. However, this rapid economic development has significantly increased energy consumption, which is a significant cause of increased carbon emissions (Sheng et al., 2014; Dong et al., 2018a; Liao and Shi, 2018).

In June 2017, as the world's second largest emitter of greenhouse gases, the USA announced its withdrawal from the "Paris Agreement" on climate change. The Trump administration held that the "Paris Agreement" would damage the USA's economy, and claimed that the agreement had "very small" effects on global climate. In contrast to that of the USA, Chinese government proposed the idea of "the community of human density and global governance". Moreover, it has stated repeatedly that China will maintain its commitment under the "Paris Agreement" to achieve carbon emission peak and reduce carbon intensity by 60–65% (Dong et al., 2018c). Combating climate change is the common responsibility of all humanity, and no country should escape this responsibility for any reason. Thus, how to effectively reduce carbon emissions without causing a great impact on economic development has become the research focus of many scholars (Liu et al., 2016; Liu and Zhou, 2016; Dong et al., 2018b; Li et al., 2019).

The concept of carbon trading was put forward as early as 1997 in the "Kyoto Protocol". At present, carbon emission trading scheme (ETS) is recognized as effective means for reducing carbon emissions internationally. To realize its commitment regarding emission reduction targets, in late 2013 and early 2014 China successively established Carbon Emission Trading Pilot Scheme (ETPS) in seven provinces or cities around China, including Beijing, Shanghai, Tianjin, Chongqing, Shenzhen, Hubei and Guangdong. To date, the operation and development of all pilot schemes are in good condition, especially in Hubei Province, where ETPS operates the most successfully. The daily turnover in Hubei Province is more stable than in the other pilot areas, and the carbon trading system is relatively problem-free.

Based on the combined experiences of emission trading operations in the pilot areas, the "breakthrough" for carbon emission trading in China occurred when a carbon emission trading market for the power generation industry was officially launched nationwide at the end of 2017. Nevertheless, the nationwide scheme will still face the problems of continuous improvement and development in the future. Xue and Zhao (2018) points out that the limited experience obtained from the seven emission trading pilots could not provide adequate guidance for the national emission trading market. Thus, during the establishment of the national carbon emission trading market, a cautious attitude still needs to be maintained. Moreover, policy implementation must be supported by further research and evaluation.

In theory, the control of total carbon emissions by market forces can overcome the inefficiency of initial carbon emission permit allocations and release great potential for energy conservation and emission reduction. Meanwhile, when the economy is growing, a win-win situation is achieved in environmental protection and economic development. Porter and Linde (1995) consider that a reasonable and strict environmental policy can stimulate the "innovation offset" effect, and thus cannot only ameliorate the "compliance cost" of an economic entity, but also increase the production efficiency and competitiveness of that entity. This view is precisely the core concept of the Porter Hypothesis. However, in actual practice, it is a crucial question still to be investigated whether China's ETS can realize the win-win situation in economic

development and environmental protection, and give full play to the Porter Hypothesis regarding the scheme.

In this study, based on Porter Hypothesis theory, the concepts of environmental dividend and economic dividend proposed by Tu and Chen (2015) are examined. In the short-term and long-term compared with a traditional Command-control Scheme (CCS), the questions investigated are whether the implementation of a carbon emission trading pilot policy can lead to environmental dividend and economic dividend, and thereby realize the Porter effect.

The remainder of the paper is organized as follows. Section 2 briefly reviews the related literature. Section 3 describes the research methods and data in this study. Results and discussion are presented in Section 4. Finally, we conclude this study.

2. Literature review

Environmental protection has a close connection with economic development; however, there are two opposite views in academic circles about the actual nature of the relationship. One view holds that there are obstructive relationship between economic development and environmental protection; the other view argues that these goals supplement each other. Neoclassical economic theory suggests the former relationship. This theory supports the view that although strict environmental protection can produce a certain social benefit, it inevitably generates an extra private cost for an economic entity, hinders increases in productivity, lowers market competitiveness, and ultimately has negative effects on economic development. In its early days until 1991, this viewpoint is supported by most scholars (Jorgenson and Wilcoxon, 1990). Then, Porter (1991) puts forward a completely opposite viewpoint to the traditional neoclassical economic theory on environmental protection. He considers that reasonable but strict environmental protection could inspire innovation, improve production technology, create productivity benefits, and thereby offset the cost due to environmental governance, increase the competitive advantage of economic entities internationally. The viewpoint suggested by Porter is now called the Porter Hypothesis. In 1995, Porter and Linde explain in detail the process by which environmental protection could promote competitiveness through innovation; the explanation further improves the Porter Hypothesis. During research on the effects of environmental regulation on the competitiveness of the manufacturing industry in the USA, Jaffe et al. (1995) point out that there is not enough evidence to judge which is right or wrong — the Porter Hypothesis or neoclassical economics theory. Since the Porter Hypothesis is proposed, some scholars have raised questions about its validity (Brannlund et al., 1995; Palmer et al., 1995; Simpson and Iii, 1996), but many scholars have found evidence that verifies the hypothesis (Johnstone et al., 2010; Rexhäuser and Rammer, 2014).

The premise for achieving the Porter Hypothesis is to satisfy the assumed conditions; that is, reasonable but strict environmental protection must be implemented through the imposition of appropriate environmental protection standards and strict enforcement by government, which stimulates the economic entity to innovate. Ultimately, the costs arising from regulation are reduced and competitiveness is increased as the economic entity offsets the effects of regulation through innovation (Debnath, 2015; Ramanathan et al., 2017). The foundation for achieving Porter Hypothesis is to formulate an appropriate standard and execute it strictly, which are actions also emphasized by the Porter Hypothesis. An unreasonable standard and poor actual execution will not stimulate the economic entity to innovate actively (Debnath, 2015; Ramanathan et al., 2017).

ETS has been considered by economists to be an effective means of environmental governance (Dales, 1968). As a control measure for total carbon emissions, carbon emission trading was put forward in the "Kyoto Protocol"; later, the European Union, the USA and Canada successively established carbon emission trading markets. Compared with ETS, pollutant trading scheme were proposed earlier and applied

earlier in practice. Thus, there is a large body of literature on pollutant trading scheme. In essence, a ETS is the same as a pollutant trading scheme, with the main distinction between the two being the trading objects. Thus, the developmental experience and research regarding pollutant trading scheme can provide guidance for devising, implementing and evaluating a ETS.

The basis of an emission permit is that an environmental resource is regarded as a good in an economic context, possesses the general properties of a commodity, and belongs to the government. Accordingly, the environmental resource is a transferable asset that has economic value (Dales, 1968). Some scholars have pointed out that the most effective way to solve environmental pollution problems is to allocate emission permits based on market principles (Coase, 1960; Cropper and Oates, 1992). Previous research on pollutant trading scheme has mainly concentrated on comparisons to traditional CCS, and on examining emission reduction effects and cost-effectiveness (Montgomery, 1972). With the development and improvement of research methods, especially the application and generalization of data envelopment analysis (DEA) models in environmental economics, many scholars have applied DEA to pollutant trading scheme, investigating and analyzing the potential economic output that can be realized (Färe et al., 2014; Tu and Chen, 2015).

The Chinese government has deployed pilot schemes for carbon emission trading from the start of 2011, and launched seven carbon emission trading markets in succession at the end of 2013 and the beginning of 2014. The carbon emission trading market plan was implemented nationwide in 2017. Moreover, at the end of 2017, a breakthrough was achieved when the national carbon emission trading market for the electric power industry was launched successfully. Since the carbon emission trading pilot policy was first put forward in China, more scholars have begun to conduct research about the issue of carbon market. Empirical studies are mainly concentrated on the results of carbon emission reduction, the effects on economic output, and other related effects. In terms of emission reduction, Cheng et al. (2008) and Lee et al. (2015) argue that there are obvious differences in the emission reductions among different industrial sectors due to carbon trading. Zhang et al. (2016) simulate the scenario in which a unified national carbon market is implemented in China, and analyze the emission reductions due to the national market. Regarding economic impact, although ETS studied is in a pilot period, Tang et al. (2015), Zhang (2015) and Wang et al. (2016) use simulations to investigate the possible effects of a comprehensive carbon trading on economic output. In terms of other related effects (for example, in consideration of trade cost problems in ETS), Wang et al. (2018) research the effective coverage of a ETS. By constructing a provincial carbon trading model, Cui et al. (2014) analyze the cost reduction effect due to carbon trading. Zhu and Chen (2018) hold that research on a ETS should consider the heterogeneity of carbon traders, and should not be limited to the rational equilibrium environment. Zhu et al. find that although a ETS could promote emission reductions in enterprises, the advantage in cost reductions is not verified. Based on CGE model, Hübner et al. (2014) study the loss in GDP caused by realizing emission pledge made in the Copenhagen Accord through the implementation of a ETS in China. The results show that the GDP loss could be controlled to about 1%, in contrast to 0.28% predicted in the research of Wang and Pan (2009).

The following developments and innovations are contributed in this study. First, to date, most of empirical research has only evaluated the results of emission reduction or potential economic output; fewer have focused on the economic function and emission reduction function of ETS. Furthermore, there are few studies that have deeply analyzed the Porter effect of ETS. In this study, from the short-term (actual) and long-term (potential) perspectives, the economic function and emission reduction function of ETS are examined. Moreover, it is also explored whether the actual and potential Porter effect could be generated. Second, based on the DEA model evaluating the implementation of a pollutant trading scheme (Färe et al., 2014), a reasonable improvement of the model is achieved in this study. Considering the special situation that a

ETS was implemented only in some Chinese pilot provinces, a DEA model introducing the carbon emission trading pilot scheme is established. Third, the Porter effect of ETS is examined relative to a CCS. That is, the question of whether there exists the potential Porter effect is investigated by assuming efficient market operation and full carbon emission permit trading, as well as the optimum production technology under two environmental policies in each province. In this way, the research is able to comprehensively compare the superiority and inferiority of the two environmental policies. Thus, the results and policy implications can provide significant reference for the development of the nationwide carbon emission trading market launched in China.

3. Methodology and data

3.1. Methodology

3.1.1. Model of testing actual (short-term) Porter effect

When it comes to the question of whether a policy is effective, it is natural for us to evaluate the effects of the policy by comparing the differences of economic individuals before and after the implementation of the policy. But it is not reasonable to simply compare the differences before and after the policy. As economic individuals may be affected by factors such as time, macro economy and random disturbance at the same time, it cannot purely reflect the policy effect, but DID method can solve the problem well.

The difference-in-difference analysis is a relatively effective policy evaluation tool. Due to its simplicity, easy operation and clear logic, this type of analysis has been widely used to evaluate policy effects in the field of social sciences. To assess the net impact of the implementation of the emission trading pilot scheme policy, we divide the entire sample data into two groups. One group, the treatment group, is affected by ETS; the other group, the control group, is not affected by ETS. The gross regional domestic product (regional GDP) and carbon dioxide emissions are selected as indicators of economic individuals. Two groups of variations are obtained by the first difference before and after the ETS implementation. After the first difference, the heterogeneity of the individual not changing with time can be eliminated. Then a second difference is conducted between the two groups to eliminate the incremental change over time, and the net effect of ETS implementation is finally obtained.

In this study, the carbon emission trading pilot policy serves as a quasi-natural experiment from 2013. The level of investigation is the province, therefore, Shenzhen City is included in Guangdong Province. Six provinces are designated as the treatment group, including Shanghai, Chongqing, Beijing, Guangdong, Tianjin and Hubei. These are areas where pilot carbon emission trading schemes had been implemented. Thus, non-pilot provinces are assigned to the control group. The non-pilot period and pilot period are “before 2013”¹ and “after 2013”, respectively. The model is constructed as follows.

$$Y_{it} = \beta_0 + \delta_0 tim_t + \delta_1 prov_i + \delta_2 tim_t \cdot prov_i + v_i + v_t + \varepsilon_{it} \quad (1)$$

$$CE_{it} = \beta'_0 + \delta'_0 tim_t + \delta'_1 prov_i + \delta'_2 tim_t \cdot prov_i + v_i + v_t + \varepsilon_{it} \quad (2)$$

$$\ln Y_{it} = \beta_0 + \delta_0 tim_t + \delta_1 prov_i + \delta_2 tim_t \cdot prov_i + \beta_1 \ln K_{it} + \beta_2 \ln L_{it} + \beta_3 \ln E_{it} + v_i + v_t + \varepsilon_{it} \quad (3)$$

$$\ln CE_{it} = \beta'_0 + \delta'_0 tim_t + \delta'_1 tim_t + \delta'_2 tim_t \cdot prov_i + \beta'_1 \ln POP_{it} + \beta'_2 \ln GDP_{it} + \beta'_3 EIP_{it} + \beta'_4 coal_rate_{it} + \beta'_5 second_rate_{it} + v_i + v_t + \varepsilon_{it} \quad (4)$$

¹ Since the emission trading pilot was officially launched in late 2013 and early 2014, year 2013 is included in the non-pilot period.

In Eqs. (1) and (2), i represents the province, and t is the year. Y and CE are the gross regional domestic product (regional GDP) and carbon dioxide emissions, respectively. The dummy variable tim is the pilot policy implementation period for carbon emission permit trading, and is assigned a value one or zero for the period after and before the policy implementation, respectively. $prov$ represents the province implementing the pilot policy of carbon emission permit trading. The pilot province and non-pilot province are valued one and zero, respectively. v_i represents the fixed effect in control provinces, and v_t is the fixed effect in the control period. ε_{it} is the residual, δ_2 and δ_2' represent the effects of ETS on GDP and carbon dioxide emissions. As the non-pilot provinces primarily use CCS to regulate environmental quality, the test of the Porter effect under ETS is relative to CCS. Investigating the Porter effect means uncovering whether the environmental policy (carbon emission trading) could generate an economic dividend and environmental dividend. If the test results show that δ_2 in Eq. (1) or δ_2' in Eq. (2) is significantly different from zero, an obvious economic dividend or environmental dividend exists; therefore, in contrast to CCS, ETS can realize short-term Porter effect.

According to the neoclassical economic growth theory, the control variables, such as the capital (K), labor force (L) and energy consumption (E) are introduced in Eq. (3). The control variables, population size (POP), economic scale (GDP), energy intensity (EIP), proportion of coal consumption ($coal_rate$) and proportion of secondary industry ($second_rate$) are introduced in Eq. (4) following previous literature (Guo, 2010; Dong et al., 2015; Deng et al., 2014).

3.1.2. Model of testing potential (long-term) Porter effect

Different environmental policies can stimulate different output potential (optimal emissions) of the economic entities. The greater the potential, the more effective the corresponding environmental policies. This study investigates the two main categories of environmental policy for carbon emission reduction, i.e., the command-control scheme and the carbon emission trading scheme. In essence, CCS is a compulsory carbon emission reduction policy in which the adopted measure includes a specific emission set for various economic entities. Excessive emissions will result in large fines or closure. In contrast to CCS, ETS not only controls the total quantity of carbon emissions, but also imposes an emission constraint on individual carbon emitters. However, it is necessary to allocate an initial emission for each entity before trading begins. The implementation of ETS can solve the inefficient allocation of carbon emission permits caused by compulsory emission reduction. Each economic entity can decide on its own final emission level according to its own situation or market condition. Increases in potential economic output and reductions in potential carbon emissions in a region (or entire country) are evidence of the Porter effect due to carbon emission trading.

Based on this idea, this paper adopts the nonparametric mathematical programming technology DEA model and assumes that the market is operating effectively. Under the condition of controlling the total amount of pollutant emission, all economic entities are allowed to freely trade carbon emission rights and effectively allocate carbon emission, so as to realize the maximization of potential output. As the simulation of environmental policy should be based on the theoretical framework of an environmental production function, DEA model can satisfy the conditional constraint. Based on the theory of environmental production technology (Färe et al., 2007; Tu, 2008; Wang et al., 2008), the models for potential output under CCS and potential output under the emission trading pilot policy are constructed. Moreover, the potential output and potential carbon emission of economic entities under each corresponding environmental policy are measured and compared. The long-term Porter effect under the carbon emission trading pilot policy is also analyzed relative to CCS. By comparing different environmental policies, the long-term Porter effect of ETS is verified.

Based on the theory of environmental production function, 30 provinces and cities (except Tibet, Hong Kong, Macao and Taiwan) serve as the decision making units in China. GDP is regarded as the desirable output variable y . Carbon dioxide emissions are considered as the

undesirable output $u \in R_+^1$. The regional labor force, capital stock and energy input are selected as the input variables $x = (x_1, \dots, x_N) \in R_+^3$. It is assumed that the input and desirable output have strong disposability, while the desirable output and undesirable output have zero correlation and weak disposability. The research data are the related input-output data over the years ($t = 1, \dots, T$) in each province ($k = 1, \dots, K$). Through the model, the maximum possibility of increased output $y \in R_+^1$ is calculated. On this account, the potential output growth and scope for emission reduction under the existing carbon emission policy in all regions are evaluated.

3.1.2.1. Potential output model under the command-control scheme. To control carbon emission at a given level, an upper limit of emissions for each region is set, and the realizable output of any region is maximized without exceeding the upper emission limit. This scenario represents the essence of a compulsory emission reduction policy. Accordingly, in each region k , under a certain input and undesirable output, the maximum output in the period t is $\tilde{y}_k^t$. The DEA model implementing CCS is described by Eq. (5).

$$\begin{aligned} Y^{CC} = & \max \sum_{k=1}^K \tilde{y}_k^t \\ \text{s.t. } & \sum_{k=1}^K z_{kk'} y_k^t \geq \tilde{y}_{k'}^t, k' = 1, \dots, K \\ & \sum_{k=1}^K z_{kk'} u_k^t = u_{k'}^t, k' = 1, \dots, K \\ & \sum_{k=1}^K z_{kk'} x_{k,n}^t \leq x_{k',n}^t, n = 1, 2, 3, k' = 1, \dots, K \\ & z_{kk'} \geq 0, k = 1, \dots, K, k' = 1, \dots, K \end{aligned} \quad (5)$$

It should be noted that each province has implemented CCS in reality, and the observed data are the actual data arising from this scheme. Thus, the difference between the potential desirable output (determined by simulating CCS) and the observed data reflect the operational efficiency of the actual CCS. Moreover, each carbon emission trading pilot province has actually implemented a ETS. The observed data from these provinces are the actual data influenced by the operation of a ETS. The difference between the potential desirable output and the actual observation data from ETS reflects the comparison of economic benefits generated by the efficient operation of the actual ETS and CCS.

3.1.2.2. Potential output model under the carbon emission trading pilot policy. The carbon emission trading pilot policy means that ETS is implemented in some areas. In the model, F regions participate in the carbon emission permit trading, and the remaining ($K-F$) regions do not. In a region implementing ETS, u_k^t represents the initial carbon emission permit, and $\tilde{u}_k^t$ is the final emissions. The DEA model introducing ETS is shown as Eq. (6). In Eq. (6), the right side of the second constraint condition shows that when the gross output is maximized, the region implementing ETS can allocate the carbon emission permit freely, which is designated as a variable in the model. The combination of the second constraint condition and last constraint condition represents the scenario in which all regions are allowed free emission allocation, while the total carbon emissions after trading are less than the initial total carbon emissions.

$$\begin{aligned} Y^{TP} = & \max \sum_{k=1}^K \tilde{y}_k^t \\ \text{s.t. } & \sum_{k=1}^K z_{kk'} y_k^t \geq \tilde{y}_{k'}^t, k' = 1, \dots, K \\ & \sum_{k=1}^K z_{kk'} u_k^t = u_{k'}^t, k' = 1, \dots, F \\ & \sum_{k=1}^K z_{kk'} u_k^t = u_{k'}^t, k' = F+1, \dots, K \\ & \sum_{k=1}^K z_{kk'} x_{k,n}^t \leq x_{k',n}^t, n = 1, 2, 3, k' = 1, \dots, K \\ & z_{kk'} \geq 0, k = 1, \dots, K, k' = 1, \dots, K \\ & \sum_{k=1}^F \tilde{u}_k^t \leq \sum_{k=1}^F u_k^t \end{aligned} \quad (6)$$

It is worth pointing out that during the construction of a DEA model for carbon emission permits or pollution rights, some scholars do not apply the equality sign, but instead use the inequality sign for the constraint condition of carbon emission or pollutant emission (Tu and Chen, 2015). This choice actually treats the carbon emissions or pollutant as an input, which can give rise to inaccurate projections due to the incorrect classification of carbon emissions. To guarantee the accuracy of estimates in this study, the equality sign is adopted in Eqs. (5) and (6).

3.1.2.3. Potential Porter effect of pilot policy for carbon emission permit trading. Referring the definition of potential economic dividend and environmental dividend given by Tu and Chen (2015), through Eqs. (5) and (6) the potential economic output under the implementation of CCS in all provinces is calculated to be $y_k^{CC,t} = \tilde{y}_k^t$, and the potential economic output and carbon emission under the pilot policy of carbon emission permit trading are $y_k^{TP,t} = \tilde{y}_k^t$ and $u_k^{TP,t} = \tilde{u}_k^t$, respectively. Thus, the potential economic dividend (PECD) and potential environmental dividend (PEND) under the pilot policy of carbon emission permit trading in each province and throughout the country can be calculated.

$$PECD_k = y_k^{TP,t} - y_k^{CC,t}, PEND_k = u_k^t - u_k^{TP,t} \quad (7)$$

$$PECD = \sum_k^F y_k^{TP,t} - \sum_k^F y_k^{CC,t}, PEND = \sum_k^F u_k^t - \sum_k^F u_k^{TP,t} \quad (8)$$

The changes in environmental dividend and economic dividend also are observed to verify the Porter effect of ETS. If the potential economic dividend and environmental dividend exist simultaneously, the frontier production technologies under the two contrasting environmental policies are heterogeneous. Compared to CCS, ETS can realize the long-term Porter effect. On the contrary, if the economic dividend and environmental dividend do not exist, the frontier production technologies under the two environmental policies are homogeneous, and ETS cannot realize the Porter effect in the long-term. Such a scenario requires the existing production technology to change through technological innovation or the introduction of advanced technology from elsewhere. Whether the new Porter effect is realized is a question that remains for further investigation.

3.2. Data

In China, both the carbon emission trading pilot policy and command-control scheme focus on the provincial level. Thus, GDP is selected as the economic output for the DEA models. Data for analysis are obtained for 30 provinces in China for during 2006–2015. The various data are mainly derived from China Statistical Yearbook (CNBSa, 2007–2016), China Energy Statistical Yearbook (CNBSb, 2008–2017), China Demographic and Employment Statistics Yearbook (CNBSc, 2008–2014), and Compilation of New China Statistics for 60 Years (CNBSd, 2010).

The difference-in-difference approach is employed to test whether the carbon emission trading pilot policy generates Porter effect in the short-term. For Eq. (3), the capital stock (K) serves as capital, the annual number of employees (L) is labor force, and the energy consumption (E) represents energy use in each province. For Eq. (4), the total permanent resident population is selected as the population size (POP), and the carbon dioxide emissions (CE) are calculated according to the method adopted in the research of Dong et al. (2014). When the DEA model is utilized to verify whether ETS realizes the long-term Porter effect, the input data are the capital stock (K), annual number of employees (L), energy consumption (E), GDP is the desirable output (Y) in each province, and carbon dioxide emissions are the undesirable output (u).

3.3. Framework of the whole research

Fig. 1 shows the research framework used in this study. As the national emission trading market is gradually constructed, it requires more thorough and comprehensive examination whether ETS can give full play to Porter's hypothesis effect and realize a win-win situation in the economy and environment. In view of current research gaps, this study focuses on the economic development function and emission reduction function of ETS simultaneously. Based on Porter's hypothesis theory, this study employs the difference-in-difference approach and the DEA models introducing the carbon emission trading pilot scheme to examine whether the carbon emission trading pilot policy can generate economic dividend and environmental dividend from short-term (real) and long-term (potential) perspectives. Meanwhile, by analogy with the environmental policy of CCS, when the optimized production technology is reached in each province under two environmental policies, the question of whether the potential Porter effect exists is explored. This provides a theoretical basis for comparing the advantages and disadvantages between two environmental policies, that is, the carbon emission trading scheme and command-control scheme. Moreover, it also has significant policy implications for the development of national carbon emission trading market launched in China.

4. Results and discussion

4.1. Test results of the short-term Porter effect under the carbon emission trading pilot policy

To more accurately utilize Eqs. (1)–(4), least squares dummy variables (LSDV) are performed, which results in more accurate parameter estimates compared to those produced by the fixed effect method. Table 1 lists the estimates of GDP and carbon dioxide emissions.

The analysis using Eqs. (1) and (3) show that there is a certain negative impact from the implementation of ETS on regional GDP when the control variables are not introduced; however, the impact is not significant. This finding is consistent with the practical situation. In practice, many imperfect conditions exist at the beginning of a carbon emission trading market; thus, an economic dividend cannot be generated by carbon emission trading. Nevertheless, when the other important control variables are introduced, the results change considerably. ETS exhibits a positive effect on economic growth, though the effect is not significant. This result indicates that under the circumstances of controlling the other variables, ETS has not significantly increased regional GDP to achieve the economic dividend. Nevertheless, ETS has a positive effect on economic growth, showing that the scheme plays a certain active role in economic growth.

The analysis through Eqs. (2) and (4) specifies that whether the control variables are introduced or not, ETS generates obvious emission reductions. Furthermore, when compared to the situation without the introduction of the control variables, the situation after the introduction of the control variables reveals a decrease in the absolute value of coefficient of interaction effect on carbon dioxide emissions. This result shows that under the circumstances of controlling the other variables, there appears a certain decline in the emission reduction under ETS, but the change in the effect coefficient is more remarkable. At present, there is little literature on the true policy effect of ETS. In most of the research on the policy effect of pollutant trading scheme, the results indicate that pollutant trading scheme has not yet generated obvious actual emission reductions. In this study, the results specify that the implementation of ETS can generate significant emission reduction. Theoretically, this proves the validity of ETS and the necessity to establish a national emission trading market.

However, the comprehensive analysis also shows that ETS does not significantly increase regional GDP in the short term; consequently, an economic dividend could not be generated. Nevertheless, after the introduction of the control variables, ETS has a positive influence on

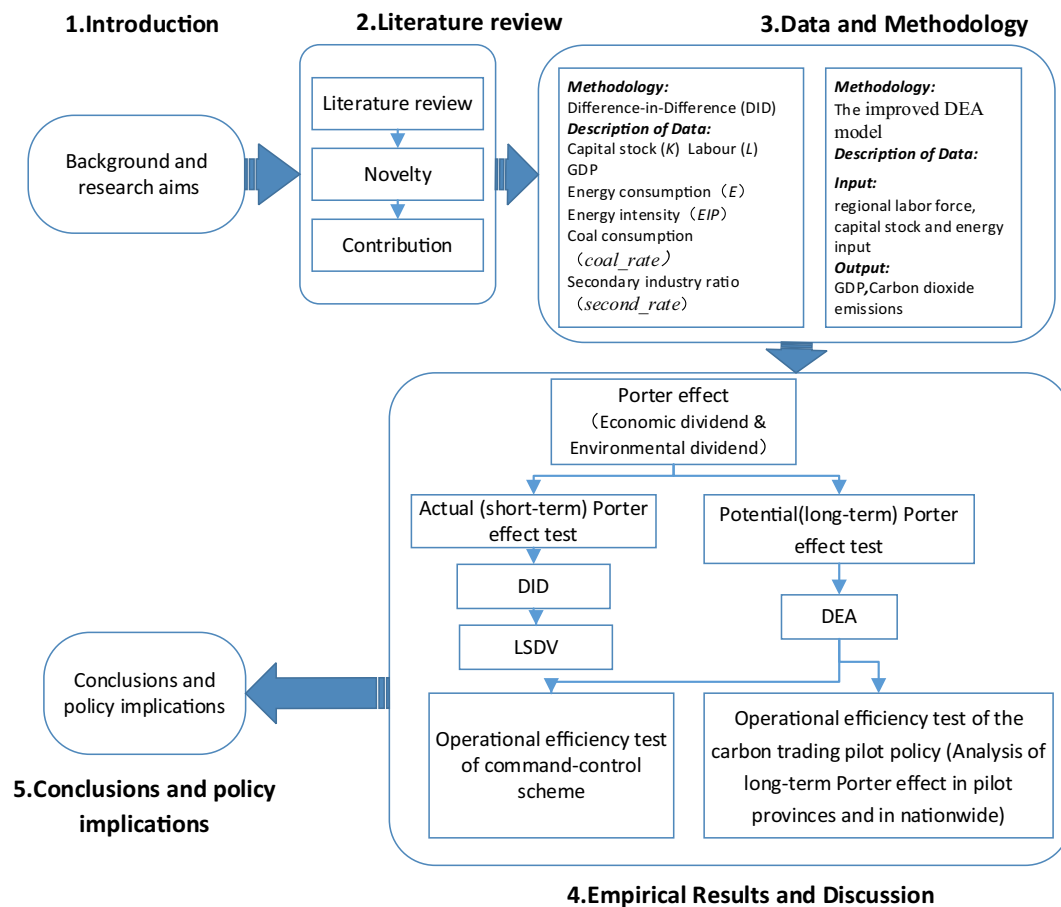


Fig. 1. Research framework.

economic development. In terms of an environmental dividend, ETS can produce a remarkable environmental dividend in the short run. This illustrates that ETS is superior to CCS in terms of carbon emission reduction. ETS produces a remarkable environmental dividend, but has not yet realized the economic dividend in the short-term. Therefore, ETS has not generated a short-term Porter effect.

4.2. Test results of the long-term Porter effect under carbon emission trading pilot policy

4.2.1. Operational efficiency test of command-control scheme

Because the test of the Porter effect under the carbon emission trading pilot policy is made relative to the command-control scheme, the operational efficiency of CCS is explored before testing whether the

carbon emission trading pilot policy could realize the long-term Porter effect. According to Eq. (5), the potential desirable outputs produced in different provinces under CCS are simulated and then compared with the observed data. Thus, the operational efficiency of CCS is obtained (see Fig. 2).

Although China's earliest carbon emission reduction target was pledged in the 2009 United Nations Climate Change Conference in Copenhagen (Dong et al., 2018c), a principle of "energy saving and emission reduction" had been included in China's development plan before 2009. A clear and rigid constraint index related to emission reduction was put forward in the "11th Five-year" Plan. The specific requirements for energy saving and emission reduction are numerous. During the period of the "11th Five-year" Plan, the energy intensity (energy consumption per unit GDP) was to be reduced by 20%, and the total main pollutant discharge was to be decreased by 10%. Finally, the energy saving and emission reduction targets were apportioned to each province, and then to the enterprise level at a micro-scale. To achieve the emission reduction target formulated in the "11th Five-year" Plan, some regions implemented extreme measures such as switching off power and limiting electricity supply. These actions directly impacted the production efficiency of economic entities and the daily lives of citizens, and thereby negatively influenced economic development in the whole region. A different approach from that in the "11th Five-year" Plan had been taken in the previous "10th Five-year" Plan. In the earlier plan, no "energy conservation and emission reduction" targets were proposed and there were no rigid constraints such as the index for emission reduction.

Based on this background, simulations of the contrasting scenarios (see Fig. 2) show that the economic loss rate decreases continuously. One possible reason for this outcome is the progress made in improving

Table 1
Regression results of the overall test in the pilot period.

Variables	Model (1)	Model (3)	Model (2)	Model (4)
<i>period</i>	0.4641*** (0.0167)	0.0758*** (0.0272)	0.2188*** (0.0347)	−0.0271** (0.0121)
<i>tim · prov</i>	−0.0104 (0.0558)	0.0445 (0.0369)	−0.1348** (0.0524)	−0.0699*** (0.0197)
Control variables	No	Yes	No	Yes
Observations	300	300	300	300
R-squared	0.9289	0.9931	0.9486	0.9961

Note: The figures in parentheses are the robust standard deviation.

*** Denote the significant level at 1%.

** Denote the significant level at 5%.

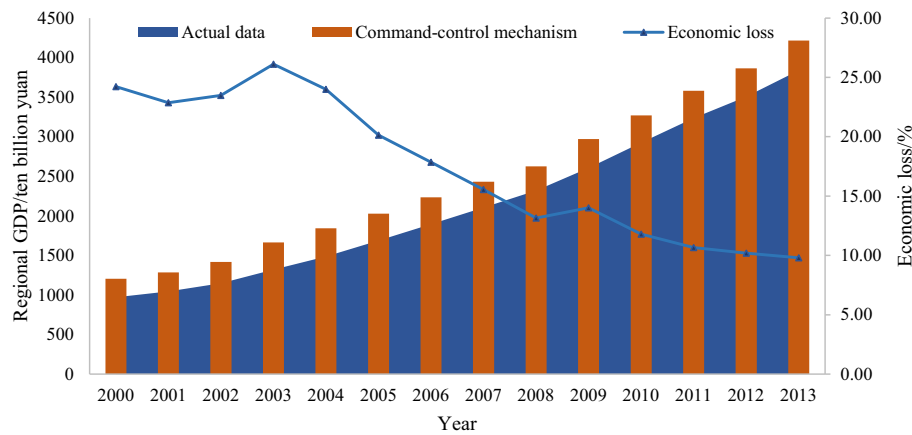


Fig. 2. Comparison of regional GDP changes under command-control mechanism simulation.

production technology in China. On the other hand, economic performance may have benefitted from the policy effect generated by CCS. The periods of the “10th Five-year” Plan (2000–2005) and “11th Five-year” Plan (2006–2010) are analyzed individually. During the period of the “10th Five-year” Plan, the observed data are the regional GDP generated in absence of CCS. In this period, the data are not affected by any policy. Thus, if CCS had been implemented and the policy implementation had been effective, the potential economic output would have been increased greatly. The actual data during the following period of the “11th Five-year” Plan describe the regional GDP generated in the presence of CCS. Therefore, the observed data are impacted by the command-control type of environmental policy in this period. The difference between the simulated data and actual data specifies the implementation efficiency of CCS.

By contrast, we can find CCS has positive effects on the improvement of economic output. Clearly, some extreme practices of CCS may have decreased the production efficiency of enterprises. However, when appropriate and reasonable measures are taken, CCS motivates many enterprises to implement innovative technology and improve production technology, thereby benefiting the enterprises. The model simulations indicate that during the period of the “10th Five-year” Plan (2000–2005), if CCS had been implemented and effectively operated, the average increment of potential output (i.e., regional GDP) would be 23.46%.

Since its inclusion in the “11th Five-year” Plan, the command-control type of environmental policy with respect to “energy saving and emission reduction” has been implemented. Theoretically, the desirable output simulated by the DEA model of CCS should be similar to the actual desirable output. Nevertheless, as Fig. 2 shows, there is a discernable difference between the output achieved under the simulated CCS and the actual economic output. This difference specifies that implementation of CCS during this period is not completely successful. Partly, the lack of success is due to an obvious deficiency in the efficiency of CCS. According to the model simulation, during 2006–2013 at national scale, the average implementation efficiency of CCS reaches only 87.13% of the theoretical efficiency. Because the undesirable output under CCS remains unchanged, the efficiency loss represents sacrificing the average potential economic output of 12.87%. Rather, in comparison to that in years not influenced by CCS, the economic loss rate declines significantly after the implementation of CCS. This occurred mainly because CCS promoted enterprises to innovate in the technology they employed and to improve production efficiency to some extent, thereby realizing an economic dividend.

From a national perspective, there appears a certain economic efficiency loss after the implementation of an energy saving and emission

reduction policy. Specifically, from the perspective of each interprovincial region, the policy implementation efficiency² in most of the developed regions is 100%. The more developed is the region, the higher is the policy implementation efficiency, and the smaller is the range over which potential output decreases (see Fig. 3). From a provincial perspective, the policy implementation efficiency in each province is uneven. Comparatively speaking, the policy implementation efficiency is higher in developed regions, and lower in relatively underdeveloped regions, where the policy fails to achieve even 50% of the desired effect. However, it is possible that some developed regions transferred some high energy-consuming enterprises to other regions, leading to higher policy implementation efficiency in the developed regions, but lower efficiency in the other regions.

Taking Beijing as an example, after the “11th Five-year” Plan, the average policy implementation efficiency of energy saving and emission reduction became 100%; in contrast, the policy implementation efficiency reaches only 67.68% in Hebei during the same period (see Fig. 3). This shows that during the implementation process of energy saving and emission reduction policy in Beijing, the economic development was not damaged. Nevertheless, it cannot be ignored that because Hebei borders Beijing, most of the pollution-intensive enterprises in Beijing migrated to Hebei, leaving mainly low-energy consuming enterprises in Beijing. Thus, there is a great difference in the policy implementation efficiency between Beijing and Hebei. Fig. 3 also presents that some of the “special” provinces (such as Shanxi and Inner Mongolia) are high energy-consuming provinces, and are also the major provinces of carbon emissions in China. However, since the “11th Five-year” Plan, many coal mining industries have been amalgamated and adjusted their business structure continuously and significantly. Thus, under the constraint of energy saving and emission reduction policy, the potential economic output has not been influenced.

In the western region (for example, Qinghai, Gansu, Ningxia and Guizhou), the implementation efficiency of CCS is lower, and do not achieve even 50% of the desired effect. This result illustrates that economic development is more dependent on the high energy-consuming enterprises in these regions. When the energy saving and emission reduction policy was implemented in high energy-consuming enterprises, because of the difficulty of transforming, some enterprises possibly adopted extreme means to realize the required energy saving and emission reduction. The extreme actions resulted in

² The policy implementation efficiency is calculated through dividing the actual output by the maximum potential output of each province under the command-control scheme.

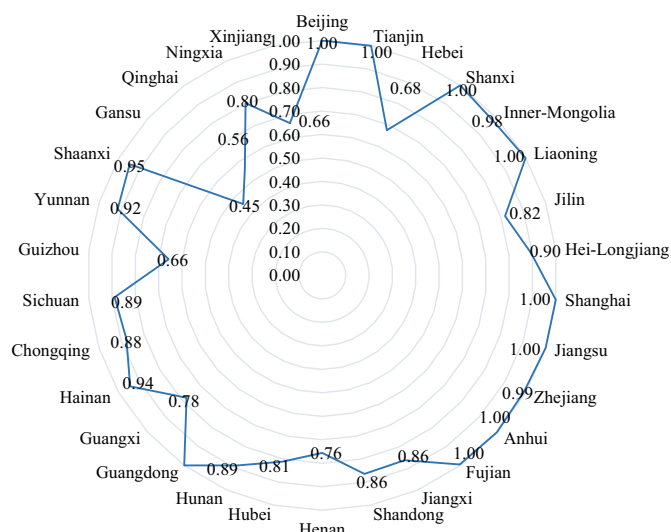


Fig. 3. Average efficiency under command-control scheme at province level during 2006–2013.

lower production efficiency for some economic entities, and thereby affected the economic output in the region.

4.2.2. Operational efficiency test of the carbon emission trading pilot policy

The results presented earlier show that there is no short-term Porter effect attributable to ETS. Moreover, the results prove that CCS can increase the potential economic output, perhaps by motivating enterprises to invent or adopt innovative technology and improve production technology, thereby generating additional economic benefit. The results seem to be only partially supportive of ETS, which can generate more significant results in terms of emission reductions than CCS, but does not significantly promote economic growth.

However, the practical experience internationally indicates that the implementation effect of environmental policy needs time to develop. This observation is reflected in the annual evaluation of the short-term Porter effect discussed above. The carbon emission trading pilot policy does not realize the short-term Porter effect compared to CCS mainly because implementation of the latter took a long period of time. To determine which of the two environmental policies is superior, the comparison must be made under circumstances that give full play to both policies. The discussion on whether ETS can realize a long-term Porter effect should be based on whether the two environmental policies are both effective. In the subsequent analysis described below, it is supposed that the emission trading market operates effectively. Then the existence of the Porter effect is investigated under the existing level of production technology achieved in each province.

4.2.2.1. Analysis of long-term Porter effect in pilot provinces. According to the DEA model constructed for the two environmental policies of command-control scheme and carbon emission trading pilot scheme, through Eqs. (7) and (8) the average potential economic dividend and

potential environmental dividend generated in carbon emission trading pilot provinces during 2014–2015 are calculated (see Table 2).

As shown in Table 2, the observed data are the actual data produced under ETS. The comparative analysis on the actual production output and the simulated production output (generated by the assumed effective operation of ETS) specifies that the simulated data are larger than the observed data, and indicates that ETS in the trading pilot did not, in fact, operate effectively. By simulating the effective operation of ETS, the annual average economic output in the trading pilot potentially increases by 339.3 billion yuan. Furthermore, the comparative analysis of actual product output and the output from the simulated CCS shows that the output increases under the simulated CCS (which operates effectively and completely). Thus, when CCS achieves completely efficient operation, it is superior to ETS that does not achieve efficient operation. When the two environmental policies both operate effectively, the increased output under ETS is higher than that of CCS. Therefore, when ETS achieves efficient operation, it is superior to CCS even when the latter is implemented effectively.

Table 2 presents the potential economic dividend and potential environmental dividend in the pilot provinces due to ETS. In terms of potential economic dividend, 15.9 billion yuan and 33.0 billion yuan of economic dividend could be released potentially in 2014 and 2015, respectively. The average potential economic growth is 0.23% during 2014–2015. In 2014, the potential environmental dividend is 121.02 million metric tons of emission reduction, and the potential emission reduction reaches 8.45%. In 2015, the potential environmental dividend is 102.68 million metric tons of emission reduction, and the potential emission reduction is 7.15%. During 2014–2015, the average emission reducing potential is 7.80%. These results verify the existence of a long-term Porter effect under ETS. The results also indicates that the pilot provinces can achieve economic dividend through the carbon emission trading pilot policy, and realize greater emission reduction in the long term by improving environmental technical efficiency. The improved technical efficiency can be achieved by adjusting the energy mix and increasing the energy utilization rate.

4.2.2.2. Analysis of the long-term Porter effect under the comprehensive implementation of ETS. The carbon emission trading pilot scheme can bring a certain potential economic dividend and potential environmental dividend for the pilot provinces. The national carbon emission trading market was officially launched at the end of 2017. For the national carbon emission trading market, further analysis of whether the comprehensive implementation of carbon emission permit trading policy can bring larger economic dividend and environmental dividend has a certain urgency and significance. Thus, a new scenario is simulated that assumes the carbon emission permit trading policy had been implemented comprehensively in the pilot period during 2014–2015. The simulation helps assess whether greater potential economic dividend and potential environmental dividend (and a more obvious Porter effect) could be generated than that produced by the existing carbon emission trading pilot policy.

Table 3 shows that in contrast to the current emission trading pilot scheme, the economic growth potential after implementing national ETS increases significantly. After national ETS is realized, the annual

Table 2
Potential dividends in six provinces under pilot carbon emission trading scheme during 2014–2015.

Year	Actual situation		CCS	ETS		Potential economic dividend	Economic growth potential	Potential environmental dividend	Emission reduction potential
	Output	Emissions	Output	Output	Emissions				
2014	976.9996	1432.2227	1002.1112	1003.7003	1311.2077	1.5891	0.1586	121.0150	8.4495
2015	1105.1780	1435.2560	1143.0342	1146.3373	1332.5758	3.3031	0.2890	102.6802	7.1541

^a The unit of output and potential economic dividend is ten billion yuan; the unit of emissions and potential environmental dividend is million tons; the unit of economic growth potential and emission reduction potential is %.

^b Economic growth potential = (Potential economic dividend / output under command-control scheme) * 100; emission reduction potential = (Potential environmental dividend / Emissions in actual situation) * 100.

Table 3

Potential dividends under full implementation of carbon emission trading policy in China during 2014–2015.

Year	Actual situation		CCS	ETS		Potential economic dividend	Economic growth potential	Potential environmental dividend	Emission reduction potential
	Output	Emissions	Output	Output	Emissions				
2014	3925.9213	10,204.3522	4361.9315	4675.6020	7100.5591	313.6705	7.1911	3103.7931	30.4164
2015	4480.6239	10,200.1636	4990.5940	5322.2425	6200.8466	331.6485	6.6455	3999.3170	39.2084

average GDP can potentially increase by 7956.5 billion yuan. These results provide additional verification that when the two environmental policies both operate effectively, ETS is superior to CCS. Implementation of ETS can greatly improve the growth potential in each region, specially, simulated economic dividend reaches 3136.7 and 3316.5 billion yuan are released in 2014 and 2015, respectively. During 2014–2015, the average economic growth potential is 6.90%, which is larger than 0.23% under the pilot policy of carbon emission permit trading. Thus, comprehensive implementation of the carbon emission trading policy can promote technical improvements in some non-pilot provinces, and thereby release more economic dividend. Moreover, the promotion is sustainable. In terms of the environmental dividend, if ETS is implemented nationwide, the potential environmental dividend will rise year by year. In 2014, the potential environmental dividend is 3103.79 million metric tons of emission reduction, and the emission reducing potential reaches 30.42%. In 2015, the potential environmental dividend is 3999.32 million metric tons of emission reduction, and the emission reducing potential is 39.21%. During 2014–2015, the average emission reducing potential is 34.81%, which is much larger than the 7.80% projected under the emission trading pilot policy.

To further compare the difference in dividends achievable between the two environmental policies, the average potential dividends in each province during 2014–2015 are calculated for a scenario in which the emission trading policy is comprehensively implemented

(see Figs. 4–7). Analysis of the variations among trading pilot provinces specifies that Beijing, Tianjin, Shanghai and Guangdong are still in the production frontier. Compared to the pilot implementation of the carbon emission trading policy, full national implementation reduces the simulated economic dividend and environmental dividend in Hubei. In comparison to its achievement under the emission trading pilot scheme, Chongqing can obtain the environmental dividend from national ETS. An analysis of non-pilot provinces shows that a certain economic dividend and environmental dividend are generated in most non-pilot regions. This illustrates that the non-pilot regions can release larger economic dividend under the comprehensive implementation of the emission trading policy. The results are consistent with the basic objective of emission trading. That is, the implementation of carbon emission permit trading can solve the inefficiency in the allocation of carbon emission permits under compulsory emission reduction. Furthermore, the increased production output associated with additional emissions from economic entities having high production efficiency is much larger than the reduced output from economic entities with low efficiency. Therefore, the total economic dividend of carbon emission permit trading can also be obtained in the whole industry or region.

Figs. 4 and 6 indicate that under the two environmental policies, varying degrees of potential improvement in the regional GDP exist in most provinces. Due to the implementation of national ETS, the annual average growth of regional GDP is much higher than that under CCS. The

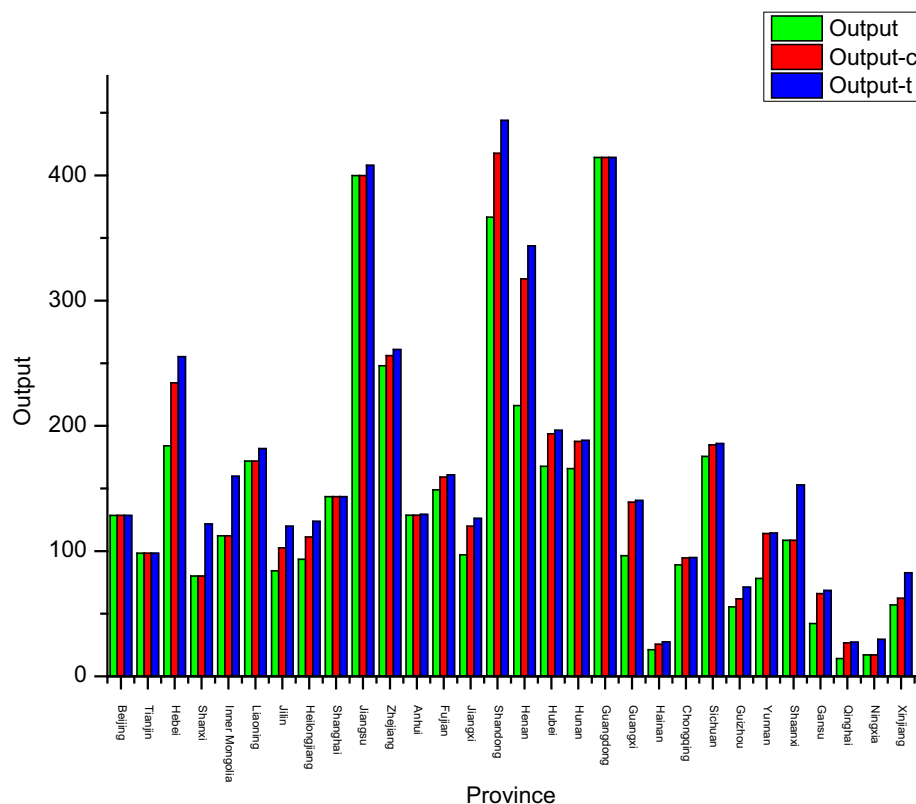


Fig. 4. Comparison of actual output, output under CCS, and output under ETS. Note: Output indicates actual output, Output-c represents output under CCS, and Output-t denotes output under ETS. Output value is the mean during 2014–2015.

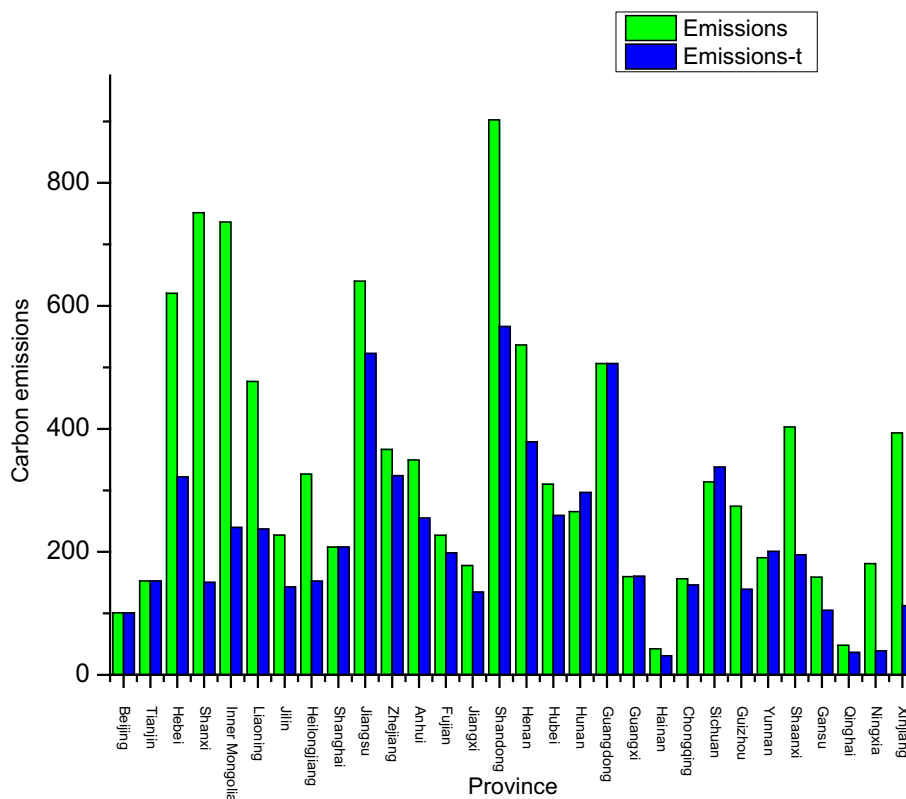


Fig. 5. Comparison of actual carbon emissions and carbon emissions under ETS.

regional GDP can be increased potentially by 7956.5 billion yuan under emission trading. Relative to CCS, this performance verifies that the implementation of national ETS can greatly improve regional GDP.

Furthermore, the economic dividend can be realized from emission trading in most provinces. During 2014–2015, the average potential

economic growth is 6.90%. On a regional basis, the potential economic output rises by 2.91% in the eastern region, by 11.53% in the central region, by 12.41% in the western region, and 6.90% nationwide. These results show that the implementation of national ETS can generate extra economic dividend by selling the carbon emission permit in the central

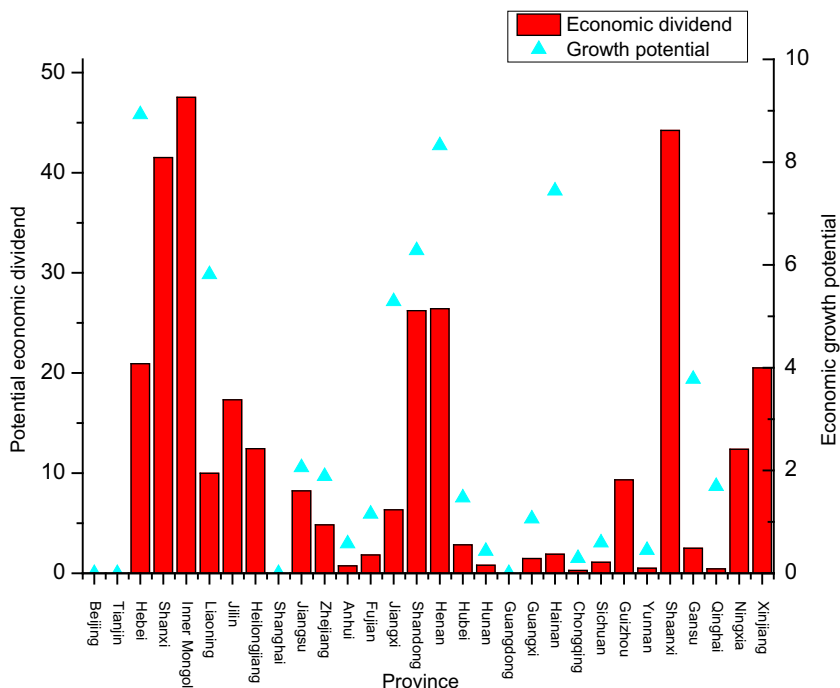


Fig. 6. Potential economic dividend and Economic growth potential.

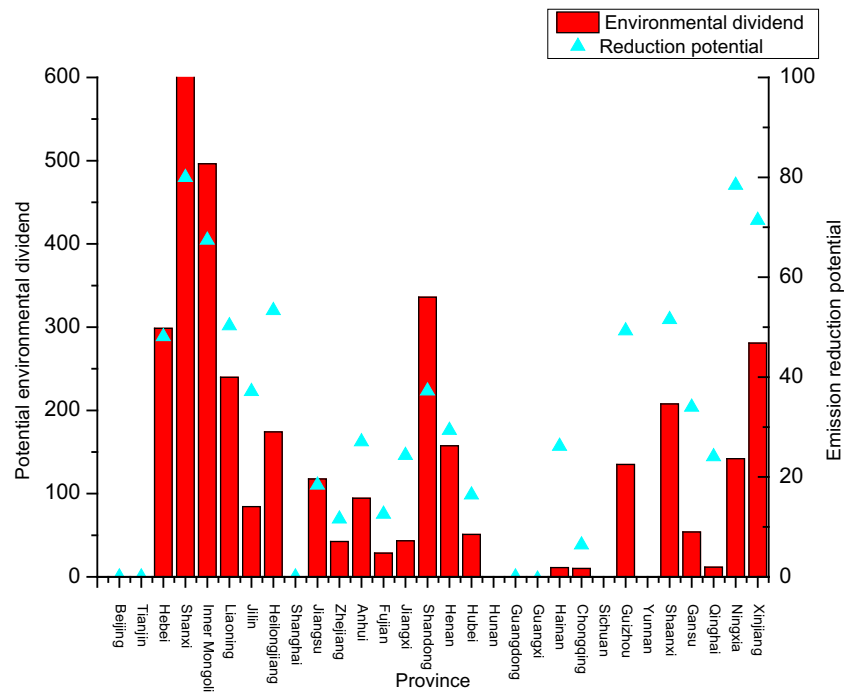


Fig. 7. Potential environmental dividend and Emission reduction potential.

and western region. In the eastern region, the output can be increased through additional carbon emission permits, thereby acquiring a larger economic dividend. Meanwhile, the improved proportion of potential economic output in the central region and western region is higher than for the national level, while in the eastern region it is lower than for the national level. This indicates that in comparison to results in the eastern region, the implementation of ETS has greater effects on, and can better drive the economic development in mid-west China. Considering the environmental dividend, the maximum emission reducing potential (79.99%) exists in Shanxi Province, where the dividend of environmental emission reduction reaches 601.04 million metric tons, followed by that in Inner Mongolia (496.37 million metric tons) and Shandong (336.01 million metric tons).

Figs. 5 and 7 specify that the carbon emissions increase in Hunan, Guangxi, Sichuan and Yunnan, illustrating that these four provinces are in the period of increasing economic development. These provinces require additional carbon emissions to boost economic development, and reflect the positive effect of the market function of ETS. From the regional perspective, the potential environmental dividend increases by 24.38% in the eastern region, by 45.40% in the central region, and by 38.07% in the western region. The average is 34.81% nationwide. These results indicate that through carbon emission permit trading, additional carbon emission reduction can be realized in the eastern, central and western regions. Similarly, carbon emission reductions in the central and western regions are higher than at the national level, but the reductions are lower than the national average in the eastern region. These results indicate that the implementation effect of ETS on the environment in the central and western regions is greater than in the eastern region. Moreover, the implementation effect can better promote the realization of carbon emission reductions in the central and western region than in the eastern region. From a different perspective, these results show that there exists a more severe inefficiency problem in the allocation of the carbon emission trading market in China at present. The full implementation of carbon emission trading can solve the initial mismatch of issuing carbon emission permits. This will also be an important way to explore the new economic growth path under the constraint of environmentally sustainable energy use in China.

5. Conclusions and policy implications

5.1. Conclusions

The national carbon emission trading market for the electric power industry, launched officially at the end of 2017, is a breakthrough in China's efforts to reduce carbon emissions. Using the difference-in-difference method and DEA models introducing the carbon emission trading pilot scheme, it is investigated whether the short-term and long-term Porter effect could be realized under the carbon emission trading pilot scheme in China. The necessity and effectiveness of the nationwide emission trading market are verified. The study results support the following conclusions.

- (1) ETS produces a remarkable environmental dividend, but has not yet realized the economic dividend in the short term. These findings differ from Wu's study that ETS can achieve the environmental dividend and release some of the economic dividend in the short term (Wu, 2018). Though the short-term Porter effect is not realized by the emission trading pilot policy. The possible reason for this failure is that the practical effect of the environmental policy is not manifested during the time period in this study. In other words, not enough time has passed since the policy was implemented. Thus, the available data could not confirm that ETS has obvious superiority in comparison to CCS.
- (2) Tests of operational efficiency under CCS confirm that although in the extreme form this scheme might lead to reductions in the production efficiency of some enterprises, the scheme could promote technological innovation and improvements in production technology, thereby bringing production benefits for other enterprises. In contrast to the scenario that any carbon emission reduction is in absence, a completely valid CCS can decrease the economic loss rate and increase the potential output.
- (3) Simulations that evaluate the long-term Porter effect confirm that when the operational efficiency of CCS is effective and

given full play, it is superior to ETS that is not operated efficiently. However, when ETS achieves efficient operation, its policy effect is better than that of CCS.

- (4) Through the emission trading pilot scheme, the long-term Porter effect is realized, and economic dividend and environmental dividend are generated. This finding is different from that in Tu and Chen's (2015) study, which suggests that the emission trading system has not achieved the Porter effect in the long term. If ETS is implemented nationwide, simulations show that it would favor energy saving and emission reduction in most provinces, thereby generating greater economic and environmental dividends for the country. Through market influences under ETS, carbon emissions are controlled on the whole, and the win-win situation of economic benefit and environmental benefit is realized.

5.2. Policy implications

This research demonstrates that ETS must be able to promote the autonomous emission reduction by carbon emitters and their innovative behavior of developing and adopting green technology. Carbon Emission trading is confirmed to be an effective environmental policy to realize carbon emission reduction, and also can ameliorate to some extent the severe inefficiency in the allocation of carbon emission permits in China. Therefore, the necessity and urgency of establishing the unified national carbon emission trading market are verified. In terms of the research results and conclusions several relevant policy suggestions are provided.

- (1) It is of great importance to rationally allocate the initial carbon emission quota for carbon emission trading parties, and improve the emission trading market system. Because the carbon markets developed in each emission trading pilot area are relatively independent, there exist great differences in the policies and regulations, quota allocation and coverage among the various carbon emission trading markets. Furthermore, because the unified national carbon emission policy is lacking in detailed guidance, the emission trading policies formulated in each pilot area are modified constantly. Thus, using actual and accurate carbon emission data, the effective implementation of an ETS needs to design the quota allocation scheme giving consideration to both equity and efficiency (Xian et al., 2018), and formulate the reasonably unified national ETS, thereby guaranteeing the operational efficiency and healthy development of carbon emission trading in China.
- (2) There is a need to increase the transparency of trade information in the carbon emission trading market, and promote the mobility of the emission trading market. ETS in China has not yet realized the Porter effect in the short-term. On the one hand, the operating duration of ETS in pilot areas is short. On the other hand, the inefficiently operating market is unable to maintain the “perfect” operation of ETS. From the perspective of developing a carbon market in each pilot region, the information disclosure of the carbon market is weak in China. This has enormous influence on the healthy development of the market. Thus, on a nationwide scale, during the establishment of a unified carbon emission trading market, it is necessary to establish a unified, public and transparent emission trading environment, and assure full disclosure of information in the market. Market manipulation and insider trading should be prevented, and direct government intervention in the market should be minimized. These actions can be promoted by taking full advantage of news media, public community and other external sources to disseminate unbiased market information.
- (3) It is vital to introduce relevant laws as soon as possible to provide legal support for the carbon emission trading market. Overall,

China currently shows the characteristics of “antecedent policy, lagged law” in the legal construction of ETS. The legal binding of policy is far lower than that of official regulations, and cannot regulate the transaction behavior of market participants. Thus, China needs laws to regulate all aspects of the emission trading market to give the market a legal basis. The formulation of a full legal system related to carbon emission trading (including laws, administrative regulations, local regulations and national government regulations) is fundamental to guaranteeing a successful carbon emission trading market. Meanwhile, several problems need to be addressed to maintain the coherence, clear legal level and internal and external coordination. Through these actions the harmony and unity of the whole legal system of carbon emission trading can be guaranteed.

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